

Introduction to Mathematics at University

Steven O'Hagan, Chris Sangwin, Wei En Tan, Cordelia Webb,
The University of Edinburgh

'Introduction to Mathematics at University' is © 2025 by Steven O'Hagan, Chris Sangwin, Wei En Tan, Cordelia Webb, The University of Edinburgh, and is licensed under [Creative Commons Attribution 4.0 International](#).

[An HTML version of this document is available online.](#)

Contents

I	Numbers	1
1	Workshops	2
1.1	Workshop 1 (Exploration)	2
1.1.1	Preparatory exercises (30 mins)	2
1.2	Workshop 2 (Reading)	3
1.2.1	Preparatory exercises (1 hour)	3
2	Summary notes	6

Block I

Numbers

1 Workshops

1.1 Workshop 1 (Exploration)

1.1.1 Preparatory exercises (30 mins)

You may already be familiar with the representation of numbers in **decimal notation** (e.g. 0.5 or 0.555...). You may also have come across numbers that can be written as **fractions of integers** (e.g. $\frac{1}{2}$ or $\frac{5}{9}$). In the Exploration workshop, we will explore the properties of each of these representations, and the connections between them.

Complete the following exercises before your first Exploration Workshop.

Exercise 1.1. Without using a calculator or computer, write each of the following real numbers as a fraction of integers.

- (a) 0.1
- (b) 0.2
- (c) 0.01
- (d) 0.11
- (e) 0.111

Exercise 1.2. Without using a calculator or computer, write each of the following real numbers in decimal notation.

- (a) $\frac{3}{10}$
- (b) $\frac{1}{5}$
- (c) $\frac{1}{10} + \frac{1}{100}$ [There are two ways you could start with this one.]
- (d) $\frac{1}{3}$

Exercise 1.3. Use whatever process you know for division to write the following fractions of integers in decimal notation.

Hint: For example, you might have learned ‘short division’ (sometimes called the ‘bus stop method’) or ‘long division’ in your previous studies.

(a) $\frac{11}{9}$

(b) $\frac{1}{7}$

[Hint: for (a) $\frac{11}{9}$ is the same as 11.0000... divided by 9.]

1.2 Workshop 2 (Reading)

1.2.1 Preparatory exercises (1 hour)

Complete the following exercises in preparation for the Reading workshop.

Divisors

Definition (Divisor). Let a and d be integers, with $d \neq 0$. We say that d **divides** a , and that d is a **divisor** (or **factor**) of a , if $\frac{a}{d}$ is an integer. We also say that a is **divisible** by d .

We shall write $d \mid a$ to mean d divides a , and $d \nmid a$ to mean that d does not divide a .

Exercise 1.4. True or false?

(a) 3 divides 12. (b) $-3 \mid 12$. (c) $12 \mid 3$. (d) $5 \mid 12$. (e) $0 \mid 5$.

Definition (Common divisor). Let a and b be integers. If $d \mid a$ and $d \mid b$ then we say d is a **common divisor** of a and b .

Definition (Greatest common divisor, gcd). Let a and b be integers, not both zero. The **greatest common divisor** of a and b , denoted by $\gcd(a, b)$, is the largest common divisor of a and b .

Exercise 1.5. Choose two integers a and b between -100 and 100 .

1. Find the common divisors of a and b .
2. What is the value of $\gcd(a, b)$?

Definition (Coprime). Let a and b be integers. We say that a and b are **coprime** if $\gcd(a, b) = 1$.

Exercise 1.6. Think of an example of two coprime numbers.

Direct Proof

Exercise 1.7. Read Section 4.2 of the [Companion Notes](#). For the following statement, identify the assumptions (objects that the statement is making a claim about) and conclusion(s) (the claim(s) that the statement is making) and try and identify where they are used in the proof.

Lemma (Linear Combination Lemma). Let a and b be integers with common divisor d . Then, for any pair of integers m and n , the integer $ma + nb$ is divisible by d .

Note that the expression $ma + nb$ above is called a **linear combination** of a and b , hence the name of the lemma.

Proof. Suppose a and b are integers with common divisor d , and let m and n be integers.

By the definition of a divisor, $\frac{a}{d}$ and $\frac{b}{d}$ are both integers. The product of two integers is an integer, and so $n \times \frac{a}{d} = \frac{na}{d}$ and $m \times \frac{b}{d} = \frac{bm}{d}$ are also integers. The sum of two integers is an integer, and so $\frac{na}{d} + \frac{mb}{d} = \frac{na+mb}{d}$ is also an integer. Hence, by the definition of a divisor, $d \mid (ma + nb)$. $\square$

Negating statements

Exercise 1.8. Read Section 3.2.1 of the [Companion Notes](#). Try to find the negation of the following statements:

1. All the peppers in the box are green.
2. If you do your homework, you'll get dessert.

Proof by Contradiction

Exercise 1.9. Read Section 4.5 in the [Companion Notes](#). Try to identify the key steps of a proof by contradiction in the following proof. If there are any steps in the proof that you are not entirely convinced by or find confusing, make a note of them to discuss in the Reading workshop.

Theorem. The number $\sqrt{2}$ is not rational.

Proof. For a contradiction, assume $\sqrt{2}$ is rational. Write $\sqrt{2} = \frac{a}{b}$, where a and b are integers so that the fraction is in simplest terms (i.e. a and b have no common factors).

Squaring both sides of $\sqrt{2} = \frac{a}{b}$ tells us that $2b^2 = a^2$.

Hence a^2 is an even number, and so a is even.

Let $a = 2k$ for some integer k . Then $2b^2 = 4k^2$ and so b^2 is even. So b is even.

Now both a and b are even. This contradicts our assumption that $\sqrt{2}$ could be written in the form $\frac{a}{b}$ with the fraction in simplest terms. Hence this assumption must be false and so $\sqrt{2}$ cannot be rational. $\square$

2 Summary notes